

Programming Assignment - 5

Complete Diagnostic Bug Analysis Report (Problems 1-5)

Problem 1 (1.c)

S.NO.	LINE NO.	ERROR DESCRIPTION	ERROR TYPE	ERROR CORRECTION
1	7	Missing semicolon after the array initialization statement of <code>marks[5]</code> .	Syntax Error	Change to <code>int marks[5] = {80,90,70,60,50};</code>
2	13	Loop bound condition <code>i <= 5</code> causes an out-of-bounds read at index 5 of the array <code>marks[5]</code> .	Array Out of Bounds	Change loop condition to <code>i < 5</code>
3	22	Format specifier <code>%d</code> used to print a <code>float</code> variable <code>average</code> . Also, integer division truncates decimal values.	Format Specifier Error / Logical Error	Change to <code>%f</code> and compute using <code>(float)total / 5</code>
4	32	Array index 5 exceeds the bounds of <code>scores[5]</code> during explicit assignment. Valid indices are 0 to 4.	Array Out of Bounds	Remove line or increase array bounds to <code>scores[6]</code>
5	43	String literal "Programming" length (11 characters + null terminator) exceeds target buffer size of <code>subject[5]</code> , causing a buffer overflow.	Buffer Overflow Error	Change declaration to <code>char subject[12];</code>
6	51	String literal "one" used as an array index for <code>attendance</code> instead of an integer type value.	Semantic Error	Change index to a valid integer constant like <code>attendance[0]</code>
7	53	Floating-point literal <code>2.5</code> used as an array index for <code>attendance</code> instead of an integer type value.	Semantic Error	Change index to a valid integer constant like <code>attendance[2]</code>
8	62	Loop bound condition <code>i <= 5</code> causes an out-of-bounds read during <code>attendance</code> array print traversal.	Array Out of Bounds	Change loop condition to <code>i < 5</code>
9	71	<code>scanf</code> statement missing the address-of operator <code>&</code> for individual array elements <code>arr[i]</code> .	Runtime Error	Change to <code>scanf("%d", &arr[i]);</code>
10	88	String literal "A" (pointer to char) assigned to character variable <code>grade[0]</code> instead of a single char literal.	Type Mismatch Error	Change to single quotes: <code>grade[0] = 'A';</code>
11	95	Assignment operator <code>=</code> used inside <code>if</code> condition instead of equality comparison operator <code>==</code> .	Logical Error	Change to <code>if(prices[2] == 300)</code>
12	104	Array variable name <code>inventory</code> used as a modifiable lvalue in direct array assignment. Arrays are non-assignable.	Semantic Error	Use a loop or <code>memcpy</code> to copy values.
13	112	Format specifier <code>%d</code> used to print a <code>float</code> variable <code>salary[0]</code> .	Format Specifier Error	Change to <code>printf("%f\n", salary[0]);</code>
14	118	Array name <code>city</code> used as a modifiable lvalue to directly assign a string literal.	Semantic Error	Change to <code>strcpy(city, "Jaipur");</code>
15	124	Loop bound condition <code>i <= 5</code> causes an out-of-bounds write at index 5 of the <code>result[5]</code> array.	Array Out of Bounds	Change loop condition to <code>i < 5</code>

Problem 2 (2.c)

S.NO.	LINE NO.	ERROR DESCRIPTION	ERROR TYPE	ERROR CORRECTION
16	16	Loop condition <code>i <= 10</code> reads index 10 from <code>numbers[10]</code> , triggering an out-of-bounds read error.	Array Out of Bounds	Change loop condition to <code>i < 10</code>
17	25	The expression <code>sum = numbers[i];</code> overwrites the <code>sum</code> value instead of calculating the accumulation.	Logical Error	Change to accumulation: <code>sum += numbers[i];</code>
32	32	Format specifier <code>%d</code> used to print the <code>float</code> variable <code>average</code> .	Format Specifier Error	Change to <code>printf("Average = %f\n", average);</code>
19	44	The condition <code>if(marks[i] < highest)</code> calculates the lowest marks instead of the highest marks.	Logical Error	Change conditional operator to <code>></code>
20	59	Explicit index assignment to <code>attendance[5]</code> exceeds the allocated bounds of size 5 array.	Array Out of Bounds	Remove the assignment or increase size.
21	65	Loop bound condition <code>i <= 5</code> causes out-of-bounds read while summing elements in <code>attendance</code> .	Array Out of Bounds	Change condition to <code>i < 5</code>
22	73	String literal <code>"ComputerScience"</code> length (15 + 1) exceeds the target buffer size of <code>subject[10]</code> .	Buffer Overflow Error	Change declaration to <code>char subject[16];</code>
23	79	Array name <code>branch</code> used as a modifiable lvalue to assign string literal directly.	Semantic Error	Change to <code>strcpy(branch, "IT");</code>
24	85	The string array <code>college[10]</code> is printed with <code>%s</code> but missing a trailing null-terminator character <code>\0</code> .	Runtime Error	Add line: <code>college[5] = '\0';</code>
25	91	String literal <code>"two"</code> used as array index for <code>fees</code> instead of an integer type value.	Semantic Error	Change index to a valid integer constant like <code>fees[0]</code>
26	93	Floating point constant <code>3.5</code> used as array index for <code>fees</code> instead of an integer type value.	Semantic Error	Change index to a valid integer constant like <code>fees[3]</code>
27	105	<code>scanf</code> statement missing the address-of operator <code>&</code> for individual array elements <code>result[i]</code> .	Runtime Error	Change to <code>scanf("%d", &result[i]);</code>
28	117	Assignment operator <code>=</code> used inside <code>if</code> condition for <code>stock[2]</code> instead of comparison <code>==</code> .	Logical Error	Change to <code>if(stock[2] == 300)</code>
29	124	Direct assignment attempted on non-assignable array lvalue name <code>backup</code> .	Semantic Error	Use loop or <code>memcpy</code> to copy values.
30	130	Format specifier <code>%d</code> used to print <code>float</code> element <code>salary[0]</code> .	Format Specifier Error	Change to <code>printf("%f\n", salary[0]);</code>
31	136	Loop bound condition <code>i <= 5</code> writes out of bounds to index 5 of <code>inventory[5]</code> .	Array Out of Bounds	Change condition to <code>i < 5</code>
32	144	String literal <code>"JaipurCity"</code> length (10 + 1) exceeds the allocated buffer size of <code>city[8]</code> .	Buffer Overflow Error	Change declaration to <code>char city[12];</code>
33	150	Direct assignment attempted on non-assignable array lvalue name <code>count</code> .	Semantic Error	Use loop or <code>memcpy</code> to copy values.

S.NO.	LINE NO.	ERROR DESCRIPTION	ERROR TYPE	ERROR CORRECTION
34	156	Missing trailing semicolon at statement <code>printf("Program Ended\n")</code> .	Syntax Error	Change to <code>printf("Program Ended\n");</code>

Problem 3 (3.c)

S.NO.	LINE NO.	ERROR DESCRIPTION	ERROR TYPE	ERROR CORRECTION
35	44	Loop bound condition <code>i <= 5</code> causes an out-of-bounds read at index 5 of <code>attendance[5]</code> .	Array Out of Bounds	Change condition to <code>i < 5</code>
36	52	String literal "Programming" length (11 + 1) exceeds the target buffer size of <code>subject[8]</code> .	Buffer Overflow Error	Change declaration to <code>char subject[12];</code>
37	58	Passing <code>&branch</code> (type <code>char (*)[10]</code>) to <code>scanf</code> instead of pointer to initial character (<code>char*</code>).	Semantic Error	Change to <code>scanf("%s", branch);</code>
38	72	Explicit element assignment to <code>fees[5]</code> exceeds the allocated bounds of size 5 array.	Array Out of Bounds	Remove statement or increase array boundaries.
39	83	Assignment operator <code>=</code> used inside <code>if</code> condition for <code>stock[2]</code> instead of comparison <code>==</code> .	Logical Error	Change to <code>if(stock[2] == 30)</code>
40	90	Direct assignment attempted on non-assignable array name <code>backup</code> .	Semantic Error	Use loop or <code>memcpy</code> to copy values.
41	96	Format specifier <code>%d</code> used to print <code>float</code> element <code>salary[0]</code> .	Format Specifier Error	Change to <code>printf("%f\n", salary[0]);</code>
42	102	<code>scanf</code> statement missing address-of operator <code>&</code> for elements <code>result[i]</code> .	Runtime Error	Change to <code>scanf("%d", &result[i]);</code>
43	114	Array name <code>city</code> used as a modifiable lvalue to directly assign string literal.	Semantic Error	Change to <code>strcpy(city, "Jaipur");</code>
44	120	String literal index "one" and floating constant 2.5 used as indices for <code>inventory</code> array.	Semantic Error	Use integer indices like <code>inventory[0]</code>
45	126	Loop bound condition <code>i <= 5</code> causes out-of-bounds read at index 5 of <code>inventory[5]</code> during print loop.	Array Out of Bounds	Change condition to <code>i < 5</code>
46	134	Character buffer <code>college[5]</code> is missing a trailing null-terminator <code>\0</code> before being printed with <code>%s</code> .	Runtime Error	Add <code>college[5] = '\0';</code> and increase size to 6.
47	140	Direct assignment attempted on non-assignable array name <code>count</code> .	Semantic Error	Use loop or <code>memcpy</code> to copy values.
48	146	String literal "Computer" length (8 + 1) exceeds target buffer size of <code>course[6]</code> .	Buffer Overflow Error	Change declaration to <code>char course[9];</code>
49	152	Loop bound condition <code>i <= 5</code> writes out-of-bounds to index 5 of <code>numbers[5]</code> .	Array Out of Bounds	Change loop condition to <code>i < 5</code>
50	162	Missing terminating semicolon at final <code>return 0</code> statement of the main function.	Syntax Error	Change to <code>return 0;</code>

Problem 4 (4.c)

S.NO.	LINE NO.	ERROR DESCRIPTION	ERROR TYPE	ERROR CORRECTION
51	22	Format specifier <code>%d</code> used to print a <code>float</code> variable <code>averageSales</code> .	Format Specifier Error	Change to <code>%f</code> inside <code>printf</code>
52	29	The condition <code>if(sales[i] < highestSale)</code> finds the lowest sales instead of the highest sale.	Logical Error	Change conditional operator to <code>></code>
53	44	Explicit out-of-bounds assignment to index 5 of <code>monthlySales[5]</code> . Valid indices are 0 to 4.	Array Out of Bounds	Remove line or increase array size.
54	48	Loop bound condition <code>i <= 5</code> creates out-of-bounds read error during print iteration.	Array Out of Bounds	Change condition to <code>i < 5</code>
55	56	String literal <code>"LaptopComputer"</code> length (14 + 1) exceeds target buffer size of <code>product[8]</code> .	Buffer Overflow Error	Change declaration to <code>char product[15];</code>
56	62	Array variable name <code>category</code> directly assigned string literal value.	Semantic Error	Change to <code>strcpy(category, "Electronics");</code>
57	68	String literal index <code>"first"</code> and float literal index <code>2.7</code> used on integer array <code>stock</code> .	Semantic Error	Use integer indices like <code>stock[0]</code>
58	78	<code>scanf</code> statement missing address-of operator <code>&</code> for element indices <code>orders[i]</code> .	Runtime Error	Change to <code>scanf("%d", &orders[i]);</code>
59	94	Assignment operator <code>=</code> used inside <code>if</code> condition for <code>revenue[2]</code> instead of comparison <code>==</code> .	Logical Error	Change to <code>if(revenue[2] == 3000)</code>
60	101	Direct assignment attempted on non-assignable array identifier name <code>backupRevenue</code> .	Semantic Error	Use loop or <code>memcpy</code> to copy elements.
61	107	Format specifier <code>%d</code> used to print <code>float</code> element <code>commission[0]</code> .	Format Specifier Error	Change to <code>printf("%f\n", commission[0]);</code>
62	113	Array variable name <code>city</code> directly assigned string literal value.	Semantic Error	Change to <code>strcpy(city, "Jaipur");</code>
63	119	Character buffer array <code>branch[5]</code> is printed with <code>%s</code> but lacks a trailing null-terminator.	Runtime Error	Add <code>branch[4] = '\0';</code> to ensure proper bounds termination.
64	125	Loop condition <code>i <= 5</code> writes out-of-bounds to index 5 of <code>employees[5]</code> array.	Array Out of Bounds	Change condition to <code>i < 5</code>
65	133	String literal <code>"Marketing"</code> length (9 + 1) exceeds the allocated target size of <code>department[6]</code> .	Buffer Overflow Error	Change declaration to <code>char department[10];</code>
66	139	Direct array assignment attempted on non-assignable array lvalue name <code>expenses</code> .	Semantic Error	Use a loop to transfer array elements individually.
67	145	Using negative index <code>-1</code> on array <code>ratings</code> causing undefined behavior out-of-bounds write.	Array Out of Bounds	Use valid indices from 0 to 4.
68	151	Format specifier <code>%s</code> passed an integer variable value <code>feedback[0]</code> instead of string address.	Runtime Error	Change to format specifier <code>%d</code>

S.NO.	LINE NO.	ERROR DESCRIPTION	ERROR TYPE	ERROR CORRECTION
69	157	Missing trailing semicolon at description output <code>printf("Report Generated Successfully\n")</code> .	Syntax Error	Change to <code>printf("Report Generated Successfully\n");</code>

Problem 5 (5.c)

S.NO.	LINE NO.	ERROR DESCRIPTION	ERROR TYPE	ERROR CORRECTION
70	22	Format specifier <code>%d</code> used to print <code>float</code> variable <code>average</code> .	Format Specifier Error	Change format specifier inside <code>printf</code> to <code>%f</code>
71	29	The condition <code>if(marks[i] < highest)</code> calculates the lowest marks instead of the highest marks.	Logical Error	Change condition to <code>if(marks[i] > highest)</code>
72	41	The condition <code>if(marks[i] > lowest)</code> calculates the highest marks instead of the lowest marks.	Logical Error	Change condition to <code>if(marks[i] < lowest)</code>
73	55	Explicit out-of-bounds assignment to index 10 of <code>attendance[10]</code> . Valid indices are 0 to 9.	Array Out of Bounds	Remove assignment statement or expand size.
74	59	Loop bound condition <code>i <= 10</code> creates out-of-bounds read error during print iteration.	Array Out of Bounds	Change loop condition check to <code>i < 10</code>
75	67	String literal <code>"DataStructures"</code> length (14 + 1) exceeds target buffer size of <code>subject[10]</code> .	Buffer Overflow Error	Change declaration size to <code>char subject[15];</code>
76	73	Array name <code>department</code> used as modifiable lvalue to directly assign string literal.	Semantic Error	Change to <code>strcpy(department, "...");</code>
77	79	String literal index <code>"two"</code> and float literal index <code>4.5</code> used on integer array <code>fees</code> .	Semantic Error	Use proper integer array indices like <code>fees[0]</code>
78	91	<code>scanf</code> statement missing address-of operator <code>&</code> for element indices <code>result[i]</code> .	Runtime Error	Change to <code>scanf("%d", &result[i]);</code>
79	107	Assignment operator <code>=</code> used inside <code>if</code> condition for <code>inventory[2]</code> instead of comparison <code>==</code> .	Logical Error	Change to <code>if(inventory[2] == 30)</code>
80	114	Direct array assignment attempted on non-assignable array lvalue name <code>backup</code> .	Semantic Error	Use loop or <code>memcpy</code> to copy array values.
81	120	Format specifier <code>%d</code> used to print <code>float</code> element <code>salary[0]</code> .	Format Specifier Error	Change to <code>printf("%f\n", salary[0]);</code>
82	126	Array variable name <code>city</code> directly assigned string literal value.	Semantic Error	Change to <code>strcpy(city, "Jaipur");</code>
83	132	Character buffer array <code>branch[5]</code> is printed with <code>%s</code> but lacks a trailing null-terminator.	Runtime Error	Add <code>branch[4] = '\0';</code> to ensure proper bounds termination.
84	138	Loop condition <code>i <= 5</code> writes out-of-bounds to index 5 of <code>stock[5]</code> array.	Array Out of Bounds	Change loop condition check to <code>i < 5</code>

S.NO.	LINE NO.	ERROR DESCRIPTION	ERROR TYPE	ERROR CORRECTION
85	146	String literal "Programming" length (11 + 1) exceeds target buffer size of course[8] .	Buffer Overflow Error	Change declaration to char course[12];
86	152	Negative array index -1 used on array ratings causing undefined memory write.	Array Out of Bounds	Use a valid integer index from 0 to 4.
87	158	Format specifier %s passed an integer variable value feedback[0] instead of string address.	Runtime Error	Change format specifier to %d
88	164	Direct array assignment attempted on non-assignable array name scores .	Semantic Error	Use an explicit loop to copy array elements.
89	170	Accessing index 5 of array temp[5] inside printf statement causes out-of-bounds read error.	Array Out of Bounds	Change target index to a valid value (0 to 4).